

SOPOCHNAYA KARGA, Russian Federation

WMO: 208710

Lat: 71.875N

Lon: 82.706E

Elev: 17

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.4	-41.1	-49.1	0.3	-43.7	-45.8	0.4	-40.9	38.6	-5.1	35.8	-0.3	7.0	90	1.102

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	67.4	60.4	63.7	58.2	60.2	55.7	61.3	66.4	58.9	63.3	56.0	59.9	9.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	74.9	63.9	56.2	67.4	62.2	53.5	61.0	58.9	27.4	66.4	25.8	63.1	23.8	60.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.0	32.4	29.3	-45.8	72.3	5.9	3.9	-50.1	75.0	-53.5	77.3	-56.9	79.4	-61.2	82.2
			-45.8	64.4	4.8	3.5	-49.2	67.0	-52.0	69.0	-54.6	71.0	-58.1	73.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	-19.3	-15.0	-5.7	3.9	19.4	39.3	51.9	47.3	39.1	20.1	-1.7	-10.6
	DBStd	27.18	14.39	15.24	16.68	13.37	9.23	7.28	7.95	6.03	6.61	10.68	13.18	14.99
	HDD50	13248	2149	1821	1728	1382	947	334	75	125	332	926	1550	1879
	HDD65	18536	2614	2241	2193	1832	1412	772	413	548	776	1391	2000	2344
	CDD50	191	0	0	0	0	0	12	132	42	5	0	0	0
	CDD65	6	0	0	0	0	0	0	6	0	0	0	0	0
	CDH74	7	0	0	0	0	0	0	7	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.1	13.7	13.4	14.1	13.3	14.3	12.2	13.4	14.6	14.7	15.5	14.0	15.4
Precipitation	PrecAvg	26.4	2.4	2.2	2.0	1.9	2.0	1.5	1.7	2.3	2.2	3.1	2.6	2.5
	PrecMax	33.5	6.0	3.9	3.3	4.5	4.2	2.9	4.2	3.9	4.5	4.7	4.4	5.0
	PrecMin	13.5	0.3	0.4	0.4	0.0	0.6	0.4	0.1	0.6	0.6	0.7	0.4	0.5
	PrecStd	4.3	1.4	1.0	0.8	0.9	0.8	0.6	1.1	0.8	0.9	0.9	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	26.1	31.2	34.2	36.0	66.0	74.9	66.1	57.0	41.0	28.9	27.3
		MCWB	26.9	25.7	30.2	32.5	34.2	58.5	63.6	59.5	53.3	39.4	28.2	26.5
	2%	DB	15.4	19.2	27.7	31.4	34.4	58.4	70.4	62.2	53.8	37.6	25.9	23.8
		MCWB	15.0	18.6	27.2	30.6	33.1	53.4	62.0	58.4	51.2	36.1	25.4	23.5
	5%	DB	5.8	13.5	23.8	28.0	33.3	53.7	66.9	58.9	51.0	35.5	21.7	18.2
		MCWB	5.3	13.0	23.1	27.5	32.3	49.8	60.2	55.4	48.7	34.4	21.0	17.4
	10%	DB	1.4	7.8	19.4	24.1	32.2	50.3	63.8	56.5	48.6	33.5	17.4	12.1
		MCWB	0.9	7.3	18.6	23.3	31.3	47.1	58.2	53.2	46.4	32.4	16.5	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	25.9	30.7	32.9	35.0	58.6	66.0	60.8	53.8	40.0	28.4	26.6
		MCDB	27.6	26.2	31.5	34.1	36.0	64.2	72.3	63.8	56.2	41.0	28.9	27.0
	2%	WB	15.0	18.7	27.1	30.5	33.4	53.4	62.9	58.6	51.3	36.3	25.2	23.4
		MCDB	15.6	19.1	27.6	31.3	34.1	56.9	68.0	61.7	53.7	37.1	25.6	23.7
	5%	WB	5.4	12.9	23.1	27.5	32.5	49.9	61.1	56.0	49.1	34.5	21.0	17.6
		MCDB	5.7	13.3	23.6	28.0	33.3	53.5	66.5	58.5	51.0	35.4	21.6	18.2
	10%	WB	1.0	7.4	18.7	23.3	31.3	47.1	58.8	53.6	46.5	32.5	16.7	11.6
		MCDB	1.6	7.9	19.4	24.0	32.2	50.4	63.5	55.8	48.4	33.6	17.5	12.1
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.7	13.8	15.3	9.2	7.2	10.2	7.8	6.1	8.0	10.4	11.7
		MCDBR	17.1	16.6	14.5	16.6	7.0	12.4	12.4	9.9	7.2	4.9	14.6	16.0
		MCWBR	16.9	16.4	14.7	16.5	6.8	9.3	8.3	7.4	6.2	4.6	14.5	15.9
	5% WB	MCDBR	17.3	15.8	14.8	16.0	7.0	11.5	11.8	8.8	7.0	5.4	14.7	15.8
		MCWBR	17.2	15.7	14.9	15.9	6.9	9.3	8.5	7.1	6.1	5.5	14.7	15.7
Clear-Sky Solar Irradiance	taub	N/A	0.219	0.291	0.314	0.338	0.349	0.373	0.358	0.308	0.215	N/A	N/A	
	taud	N/A	1.925	2.012	2.067	2.093	2.307	2.345	2.366	2.398	2.030	N/A	N/A	
	Ebn at Noon	0	123	205	249	260	263	250	238	214	124	0	0	
	Edh at Noon	0	11	25	35	39	33	31	26	18	9	0	0	
All-Sky Solar Radiation	RadAvg	0	54	471	1229	1824	1781	1536	937	412	105	0	0	
	RadStd	0	8	52	107	101	125	195	147	45	15	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	+2.86	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page